OUTLIER-ROBUST KERNEL LEARNING

Eric Scheer, Jun Simons & Anu Aryal
Rensselaer Polytechnic Institute
{scheee2,simonj11,aryala}@rpi.edu

ABSTRACT

We study robust kernel learning under the strong ϵ-contamination model, where an adversary can inspect the training set and arbitrarily corrupt an ϵ fraction of samples. Motivated by iterative thresholding gradient descent (ITGD) methods for robust learning, we develop a kernalized procedure for kernel ridge-style objectives that uses only the $(1-\epsilon)n$ samples with smallest residual loss at each iteration to reduce the influence of corrupted points. We evaluate linear and nonlinear kernel variants on synthetic and real datasets, and under varying levels and types of corruption. We find that on linear data, linear ITGD is stable across a range of corruption levels and significantly outperforms normal least squares on corrupted datasets. On nonlinear data, kernel methods offer benefits depending on choice of kernel.

1 INTRODUCTION

Modern supervised learning methods are often highly sensitive to a small number of corrupted training examples. This can be especially severe in an adversarial setting, where an adversary can see the dataset before selecting samples to modify. In such settings, even convex estimators like least-squares or kernel ridge regression can be pushed towards poor solutions.

Recent work to avoid poor model fit in these settings involves reducing the influence of training samples that appear to be inconsistent with the current model. This requires two steps: first, identify samples that best fit the model fit to avoid outliers; second, update the model using only the better fitting points.

We aim to extend this approach to kernel methods. We study an outlier robust kernel learning procedure inspired by iterative thresholding gradient descent (ITGD). Based on a kernelized objective, we perform gradient descent in the dual space while excluding examples that appear to be outliers, and further incorporate a weight vector to weigh suspicious points less.

We focus on these main questions:

1. How does robust kernel learning compare with standard baselines under various levels of contamination?
2. How do these methods perform experimentally against different types of corruption?
3. When does kernelization help, and when can it hurt, under adversarial corruption?

2 RELATED WORK

Robust statistics has long studied estimation tasks under contamination. Early work including Huber's contamination model provide statistical frameworks to model datasets that contain both clean data points and arbitrary outliers (Huber, 1964). In modern machine learning, this line of work has been extended to a broader range robust estimation tasks.

A number of regression methods exist that are designed to be robust to outliers. Alquier and Gerber develop a regression framework that minimizes Maximum Mean Discrepancy, with an estimator shown to be robust to adversarial contamination. (Alquier & Gerber, 2024)

The use of iterative thresholding for robust estimation dates back to Legendre in 1805. The simplicity of the algorithm makes it appealing to work with; however, strong theoretical guarantees regarding robustness and algorithmic tractability have only emerged somewhat recently. Bhatia et al. (2015) showed that under a linear model with dense, bounded noise $\boldsymbol{\xi}$ and unbounded, adversarial label corruption $\mathbf{b}$, ITGD returns a $\mathcal{O}\left(\frac{\|\boldsymbol{\xi}\|}{\sqrt{n}}\right)$-approximate iterate after $T = \Omega\left(\log \frac{\|\mathbf{b}\|}{\sqrt{n}\varepsilon}\right)$ iterations.

Awasthi et al. (2022) used iterative thresholding with full solves to find $\mathcal{O}\left(\sigma\epsilon \log(1/\epsilon)\right)$-approximate iterate after $T = \Omega(1/\epsilon^2)$ iterations with high probability. Here, our noise is Gaussian with mean 0 and variance σ^2 and ϵ is the fraction of labels corrupted.

More recently, Rathnashyam & Gittens (2024) used ITGD and covariate filtering (Dong et al., 2019) for learning single-neuron models with gaussian noise with similar approximation bounds. Our work attempts to extend this to Kernel Ridge Regression (KRR) which is an area that currently unexplored.

3 METHODS

3.1 PROBLEM SETUP

We assume that we are given n datapoints $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ with covariates sampled from a centered sub-Gaussian distribution with known covariance matrix Σ and sub-Gaussian norm K. We assume that our labels satisfy

$$y_i = f(\mathbf{w}_\star^\top \mathbf{x}_i) + \xi_i, \tag{1}$$

where $\xi_i \sim \mathcal{N}(0, \sigma^2)$ and f is a nonlinear activation function. We then corrupt our data under the strong ϵ-contamination model.

Definition 1 (Strong ϵ-contamination model). *Given a corruption parameter $0 \leq \epsilon < 1/2$, an adversary is allowed to inspect all samples and arbitrarily modify ϵn samples. This corrupted set of n points is then given as input to the learning algorithm.*

Note that corruption may occur in both the covariates and the labels. We denote the indices of the corrupted samples as E and the uncorrupted samples as G.

3.2 FROM REGULARIZED ERM TO KERNELIZED OPTIMIZATION

Consider the regularized empirical risk objective

$$J(\mathbf{w}) = \frac{1}{n}\sum_{i=1}^n \ell(f(\mathbf{x}_i^\top \mathbf{w}), y_i) + \frac{\lambda}{2}\|\mathbf{w}\|_2^2. \tag{2}$$

First-order optimality conditions imply that the minimizer $\mathbf{w}^\star$ must satisfy $\nabla J(\mathbf{w}^\star) = \mathbf{0}$. Therefore $\mathbf{w}^\star$ must satisfy

$$\mathbf{w}^\star = -\frac{1}{\lambda n}\sum_{i=1}^n \ell'(f(\mathbf{x}_i^\top \mathbf{w}^\star)) f'(\mathbf{x}_i^\top \mathbf{w}^\star)\mathbf{x}_i \tag{3}$$

The minimizer lies in the row space of the covariate matrix, so $\mathbf{w}^\star = X^\top \boldsymbol{\alpha}$ for some dual vector $\boldsymbol{\alpha} \in \mathbb{R}^n$. Substituting this form yields a kernelized objective involving the Gram matrix $K = XX^\top$, or more generally $K_{ij} = \kappa(\mathbf{x}_i, \mathbf{x}_j)$ for a positive semidefinite kernel κ.

For squared loss with identity link, this reduces to kernel ridge regression:

$$\min_{\boldsymbol{\alpha}\in\mathbb{R}^n} \frac{1}{n}\|K\boldsymbol{\alpha} - y\|_2^2 + \frac{\lambda}{2}\boldsymbol{\alpha}^\top K\boldsymbol{\alpha}. \tag{4}$$

Our robust optimization procedure can be described with unregularized or lightly regularized squared-loss:

$$\min_{\boldsymbol{\alpha}\in\mathbb{R}^n} \frac{1}{n}\sum_{i=1}^n \left((K\boldsymbol{\alpha})_i - y_i\right)^2. \tag{5}$$

This formulation is equivalent to linear regression when $\kappa(\mathbf{x}_i, \mathbf{x}_j) = \mathbf{x}_i^\top \mathbf{x}_j$, while also allowing nonlinear kernels such as polynomial or radial basis function (RBF) kernels.

3.3 COVARIATE FILTERING

A challenge under strong contamination is that corrupted covariates may distort the gradients even before they have large residuals. To address this, we adopt a covariate-filtering preprocessing step that produces a weight vector $\boldsymbol{\omega} \in [0, 1]^n$ as described in Dong et al. (2019).

Intuitively, these weights reduce the contribution of points with covariates that seem inconsistent with the most likely distribution of clean covariates.

These weights are incorporated directly into the gradient update step. So, this method combines two approaches that avoid model degradation:

- **covariate filtering**, which reduces the weight given to outlier covariates;
- **iterative hard residual thresholding**, a hard threshold to remove the largest-loss points at each iteration, corresponding the the points most likely to have corrupted labels.

3.4 KERNEL ITGD

Our algorithm expands upon Algorithm 1 from (Rathnashyam & Gittens, 2024) by applying hard thresholding directly the the kernel matrix K.

Definition 2 (Hard Thresholding Operator (Bhatia et al., 2015)). *For any vector $\mathbf{v} \in \mathbb{R}^n$, let $\sigma_{\mathbf{v}} \in S_n$ be a permutation that orders the entries of $\mathbf{v}$ in increasing order of magnitude, i.e., $|\mathbf{v}_{\sigma_{\mathbf{v}}(1)}| \leq |\mathbf{v}_{\sigma_{\mathbf{v}}(2)}| \leq \cdots \leq |\mathbf{v}_{\sigma_{\mathbf{v}}(n)}|$. Then, for any $k \leq n$, the hard thresholding operator is defined as the index set*

$$\mathrm{HT}(\mathbf{v}; k) = \left\{ i \in [n] : \sigma_{\mathbf{v}}^{-1}(i) > n - k \right\}, \tag{6}$$

i.e., it selects the indices corresponding to the k largest-magnitude entries of $\mathbf{v}$.

At iteration t, given a dual iterate α_t, we compute per-sample residual losses

$$r_i^{(t)} = \big((K\boldsymbol{\alpha}_t)_i - y_i\big)^2, \qquad i = 1, \ldots, n. \tag{7}$$

We then keep only the $(1 - \epsilon)n$ examples with smallest residuals:

$$S_t = \mathrm{HT}(\mathbf{r}^{(t)}, (1 - \epsilon)n), \tag{8}$$

The trimmed weighted gradient update is

$$\boldsymbol{\alpha}_{t+1} = \boldsymbol{\alpha}_t - \eta \sum_{i \in S_t} \big((K\boldsymbol{\alpha}_t)_i - y_i\big)\omega_i K_i, \tag{9}$$

where K_i denotes the ith row of the kernel matrix and η is the step size.

This update can be interpreted as: only points with reasonably small current residuals contribute to the next step, and even among these points, the covariate filter can further reduce the effect of suspicious samples.

3.5 PREDICTION

For a test point $\mathbf{x}_{\mathrm{new}}$, prediction is made by

$$\hat{y}(x_{\mathrm{new}}) = \sum_{i=1}^{n} \alpha_i^\star \kappa(\mathbf{x}_{\mathrm{new}}, \mathbf{x}_i), \tag{10}$$

In the experiments below, we focus on squared-loss regression and report test MSE.

Algorithm 1 Gradient Descent IHT on a Linear Kernel

input: Samples $D = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)\}$, Σ, ϵ, η, T, $\boldsymbol{\alpha}_0$
output: Approximate solution $\boldsymbol{\alpha} \in \mathbb{R}^n$

$D = \{(\Sigma^{-1/2}\mathbf{x}_1, y_1), \ldots, (\Sigma^{-1/2}\mathbf{x}_n, y_n)\}$
$K_{i,j} = \mathbf{x}_i^\top \mathbf{x}_j$ for all $i, j \in \{1, \ldots, n\}$
$\boldsymbol{\omega} = \textsc{CovariateFiltering}(D, \epsilon)$ ▷ Algorithm 4 of (Dong et al., 2019)
for $t = 0, \ldots, T-1$ **do**
  $r_i^{(t)} \leftarrow ((K\boldsymbol{\alpha})_i - y_i)^2$ for all $i \in \{1, \ldots, n\}$ ▷ Calculate $\mathcal{L}$ for each sample
  $S_t \leftarrow \mathrm{HT}(\mathbf{r}^{(t)}, (1-\epsilon)n)$ ▷ Hard thresholding
  $\boldsymbol{\alpha}_{t+1} \leftarrow \boldsymbol{\alpha}_t - \eta \sum_{i \in S_t} ((K\boldsymbol{\alpha})_i - y_i) \cdot \omega_i K_i$ ▷ Weighted gradient descent update
end for
return: $\boldsymbol{\alpha}_t$

4 Experiments

4.1 Experimental goals

We evaluate the following hypotheses:

1. Iterative trimming substantially improves robustness relative to ordinary least squares (OLS) trained on corrupted data.
2. Kernelizing ITGD can improve performance on nonlinear problems.
3. The effect of kernelization depends strongly on whether the chosen kernel matches the underlying target structure.

4.2 Compared methods

We compare:

- **Linear ITGD (Rathnashyam & Gittens, 2024)**: robust ITGD in the linear setting.
- **Kernel ITGD (linear)**: dual-form kernel ITGD with a linear kernel.
- **Kernel ITGD (RBF)**: kernel ITGD with an RBF kernel.
- **Huber KRR**: kernel regression with Huber loss (Huber, 1964)
- **OLS (corrupted)**: ordinary least squares trained on the contaminated dataset.
- **OLS (inlier)**: oracle baseline trained only on clean points, when available.

4.3 Data and corruption protocol

We first test with synthetic data, where we vary the corruption fraction ϵ over a wide range and report test MSE.

We study both:

- **approximately linear data**, where linear structure is well aligned with the target;
- **nonlinear data**, with a leaky-ReLU derived target that presents a better use case for kernel methods to benefit

We consider a number of types of dataset corruption

1. Setting an ϵ fraction of sample labels to zero
2. Adding random gaussian noise to an ϵ fraction of sample labels

4.4 Implementation Details

For our tests, we use a learning rate of $\eta = 0.1$ for all standard ITGD runs and a learning rate of $\eta = 0.01$ for all kernelized ITGD runs. We also run each algorithm for a maximum of 500 iterations.

Due to numerical instability, we also train using the normalizer kernel matrix where

$$(K_{norm})_{ij} = \frac{K_{ij}}{\sqrt{K_{ii}K_{jj}}} \tag{11}$$

4.5 Metrics

Our primary metric is recording the mean squared error (MSE) of each trained model on a clean dataset sampled with the ground truth distribution. We first consider test MSE over different ϵ corruption fractions and then look at training behavior at a fixed $\epsilon = 0.15$. We also measure convergence by tracking gradient norms during training at this fixed ϵ value.

5 Results

5.1 Robustness on Approximately Linear Data

We first evaluate the methods on synthetic data generated from an approximately linear model under increasing levels of corruption ϵ. We find that behavior strongly depends on the type of corruption, but linear ITGD is overall the most stable and best performing method across a wide range of ϵ.

Under additive corruption, where an ϵ fraction of samples are corrupted with large amounts of gaussian noise, corrupted OLS performs the poorly, while linear ITGD maintains performance close to the inlier-oracle baseline, and kernel ITGD also remains more stable than corrupted OLS. This indicates that filtering and thresholding is effective in preventing outlier points from dominating the gradient. Figure 1 shows this trend.

Since this task is approximately linear, the nonlinear kernel methods are not particularly suited to this.

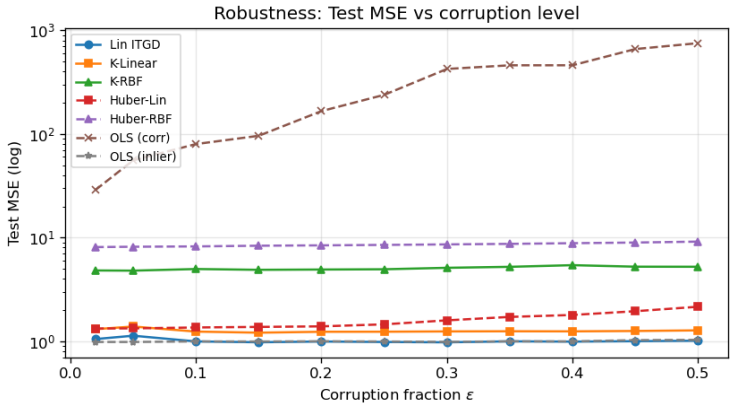

Figure 1: *Test MSE versus corruption level under additive Gaussian corruption.*

5.2 Optimization behavior

To better understand how optimization behaves, we compare gradient convergence and training MSE over the course of optimization in Figure 2. An interesting pattern is that linear ITGD converges substantially faster than all kernelized alternatives: its gradient norm decreases by several orders of magnitude and reaches essentially zero within roughly the first few hundred iterations. The kernel methods show slower decay, with gradient norms remaining larger throughout training.

This difference is also reflected in the training MSE trajectories. Linear ITGD quickly reduces training error and then stabilizes, while kernel methods change more gradually and remain at noticeably higher objective values. Among the kernel methods, the linear-kernel variant and Huberized linear KRR show the most visible decrease, whereas the RBF and polynomial variants improve more slowly.

Overall, these curves are consistent with the empirical robustness results: on approximately linear data, linear ITGD is not only the strongest method in terms of test performance, but also the easiest to optimize. The kernel are viable and do converge, but more slowly. The added flexibility of the kernelized models does not provide an advantage in the linear setting.

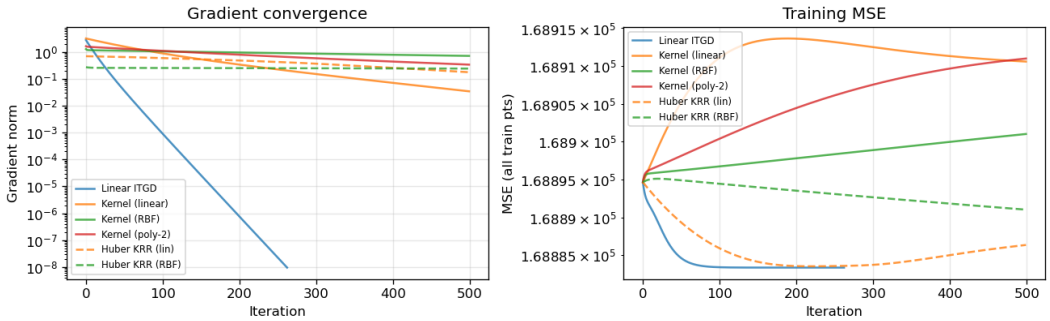

Figure 2: *Optimization behavior across method with fixed $\epsilon = 0.15$.* ***Left:*** *gradient norm versus iteration.* ***Right:*** *training MSE versus iteration. Linear ITGD converges much faster than the kernelized methods and reaches the lowest training error on this approximately linear task.*

We also examine corruption caused by setting an ϵ fraction of labels to zero. This type of corruption has less negative impact on OLS, allowing corrupted OLS to outperform several kernel baselines although it still does not match linear ITGD over low-to-moderate corruption. Figure 3 illustrates this behavior.

This experiment appears to exhibit a breaking point, where performance deteriorates and robust methods—including linear ITGD—lose their advantage.

This behavior is very different than the additive gaussian noise, as the zero-label corruption appears to be less damaging to the corrupted OLS. This emphasizes that conclusions on robustness depend not only on the fraction of corrupt samples, but also the type of corruption applied.

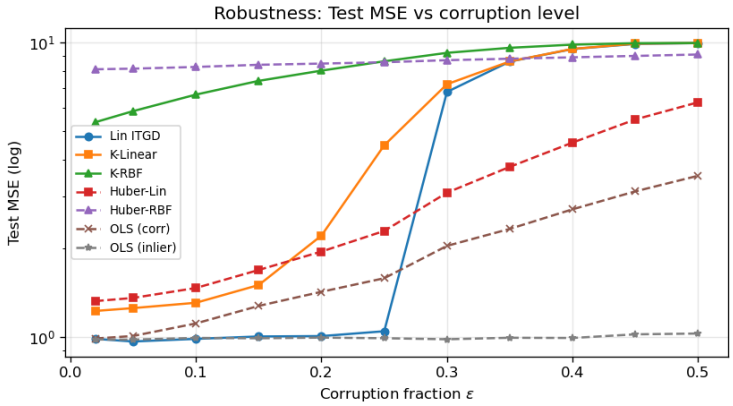

Figure 3: *Test MSE versus corruption level when an ϵ fraction of labels are set to zero. Linear ITGD performs best for low-to-moderate corruption, but a clear breakdown appears at larger ϵ.*

5.3 REAL DATASET EXPERIMENT

In addition to synthetic linear data, we also test on a social media regression dataset. Results seem consistent with the test on synthetic data: Linear ITGD performs similarly to OLS on inlier points, and K-Linear performs a bit worse. K-RBF performs poorly with the iterative filtering and down-weighting, and OLS on corrupted points does the worse. This provides evidence that the results we see are consistent when experiments are performed a real dataset.

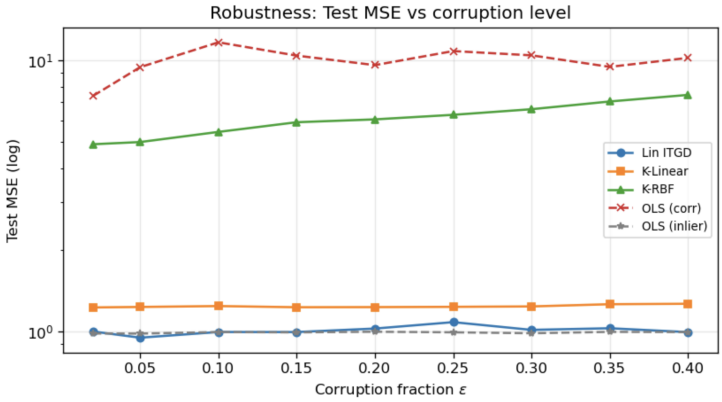

Figure 4: Test MSE vs corruption level with additive gaussian noise

5.4 NONLINEAR TARGET EXPERIMENT

Above experiments test models on linear data, for which kernel methods provide no benefit. While not supported by the theoretical portion, we further examine whether kernel ITGD is effective at robustness in a nonlinear setting, We generate data using a leaky-ReLu model, which creates a misspecification for the standard ITGD baseline, and gives the kernel methods with nonlinear kernels an opportunity to better learn this pattern.

Table 1 summarizes the outcome: as expected, linear ITGD when correctly specified with leaky-ReLU performs the best, and linear ITGD with the identity model performs much worse due to this misspecification.

K-RBF performs similarly whether evaluted with identity or leaky ReLU specification, suggesting the RBF kernel absorbs some of the unknown nonlinearity in the generated data. K-RBF also outperforms Huber-RBF, which downweights large residuals in the loss function, but does not include explicit residual thresholding and sample filtering like ITGD. It is unclear the interaction between the outlier robustness techniques and the RBF kernel, and the boost in performance may be purely due to the nonlinearity gained by the infinite dimensional transform.

The K-Linear ITGD performs worse than Linear ITGD in both cases, which is consistent with our prior results. When using a correctly specified leaky-ReLU activation function, it performs better than with the standard identity. Corrupted OLS remains worse than the robust methods in the nonlinear setting. We also test plain Kernel Ridge Regression as implemented in scikit-learn with a linear kernel, and it does the worst on the corrupted data.

Table 1: Performance summary on leaky-ReLU-generated data with additive Gaussian label corruption.

Method	Test MSE range
Lin ITGD (leaky, correct model)	$0.2355 \to 0.2459$
OLS (inlier oracle)	$0.6964 \to 0.7704$
K-RBF ITGD (identity)	$0.7956 \to 0.8914$
K-RBF ITGD (leaky)	$0.7887 \to 0.8981$
Huber-RBF (identity)	$0.9565 \to 1.1060$
Lin ITGD (identity, misspecified)	$1.1122 \to 1.1343$
K-Linear ITGD (leaky)	$1.5395 \to 1.5662$
OLS (corrupted)	$1.6164 \to 4.1233$
K-Linear ITGD (identity)	$1.7069 \to 1.7485$
KRR (corrupted, linear)	$3.1814 \to 8.7565$

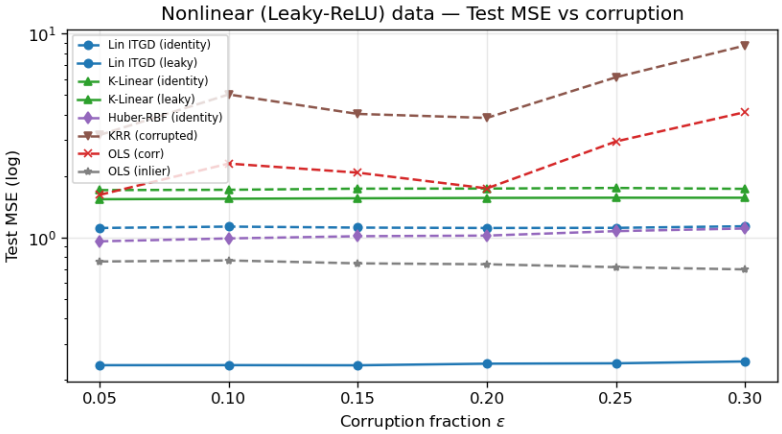

Figure 5: Performance plot for mis-specified leaky-ReLU data

6 Discussion

Our experiments show that applying covariate filtering and iterative thresholding to gradient descent for KRR has potential to be robust to outliers and adversarial noise; however, more work still needs to be done to refine these techniques to work well with arbitrary kernels where we cannot say with high confidence that rows with the highest residuals most likely correspond to the outlier samples. We perform some initial experiments with alternative kernels and observe qualitative results, but theory suggests there may be better alternative methods for nonlinear kernel functions.

Future studies could examine the existence of statistical breakpoint, likely some ϵ-level or form of corruption where model performance drastically begins to decrease, and even drops below the performance of OLS on the corrupted dataset. Additionally, alternative forms of corruption can be considered beyond additive gaussian noise or replacement with zero, especially when an adversary more carefully considers which datapoints to modify to cause maximal harm.

This work along with previous literature assumes that the corruption fraction ϵ and the ground truth covariance matrix Σ are known. In practice, this is unrealistic, so robust mean and covariance methods need to be implemented to get a good approximate of these ground truth parameters.

While working on this project, we encountered a number of issues related to implementation details. During testing, we found the gradient update rule for linear ITGD from Rathnashyam & Gittens (2024) in which we divide the gradient by $n(1-\epsilon)$ led to little/no learning. Since each covariate weight ω_i has absolute value less than $\frac{1}{n}$, this division made each gradient step essentially negligible.

REFERENCES

P. Alquier and M. Gerber. Universal robust regression via maximum mean discrepancy. *Biometrika*, 111(1):71–92, Mar 2024. ISSN 1464-3510. doi: 10.1093/biomet/asad031. URL https://doi.org/10.1093/biomet/asad031.

Pranjal Awasthi, Abhimanyu Das, Weihao Kong, and Rajat Sen. Trimmed maximum likelihood estimation for robust learning in generalized linear models, 2022. URL https://arxiv.org/abs/2206.04777.

Kush Bhatia, Prateek Jain, and Purushottam Kar. Robust regression via hard thresholding. *CoRR*, abs/1506.02428, 2015. URL http://arxiv.org/abs/1506.02428.

Yihe Dong, Samuel B. Hopkins, and Jerry Li. Quantum entropy scoring for fast robust mean estimation and improved outlier detection, 2019. URL https://arxiv.org/abs/1906.11366.

Peter J. Huber. Robust estimation of a location parameter. *Annals of Mathematical Statistics*, 35(1): 73–101, 1964. doi: 10.1214/aoms/1177703732.

Adrien-Marie Legendre. On the method of least squares. In D. E. Smith (ed.), *A Source Book in Mathematics*, pp. 576–579. Dover Publications, New York, 1805. Translated from the French (original work published 1805).

Arvind Rathnashyam and Alex Gittens. Iterative thresholding for non-linear learning in the strong ε-contamination model, 2024. URL https://arxiv.org/abs/2409.03703.